

The Configurable Logic Block (CLB) on the PIC16F13145

A precise description of what the CLB is, what feeds into it, what it is built from, what comes out of it, and how it is clocked. All facts are taken from the PIC16F13145 data sheet, chapter 29 (*Configurable Logic Block*); section numbers link to the online data sheet.

1. What a CLB is

The **Configurable Logic Block (CLB)** is a small, **FPGA-like programmable logic fabric** integrated into the microcontroller. Instead of being a fixed-function peripheral that you configure with a handful of registers (like the CLC, CWG or NCO), the CLB is a generic array of look-up tables and flip-flops whose function and interconnect are defined by a **configuration netlist**.

You do **not** program the CLB by writing to control registers directly. Instead:

1. The logic is entered through a **Configuration Interface** (a design tool, e.g. the MPLAB CLB synthesizer fed from Verilog), which produces a **netlist**.
2. The netlist is stored in **Program Flash Memory (PFM)**.
3. At run time the **NVM Scanner** (working with the CRC module) transfers the netlist from PFM into the CLB's configuration latches and LUTs.
4. Only after the scan completes is the module enabled via the **EN** bit in **CLBCON**.

See [§29.1 CLB Module Enable](#) and [§29.8 CLB Configuration](#).

The CLB exists to implement custom, low-latency hardware logic and state machines that run independently of the CPU — glue logic that would otherwise need an external CPLD/FPGA or that no dedicated on-chip peripheral provides.

2. Inputs

The CLB takes signals from three different sources, all selectable through the Configuration Interface.

2.1 CLB input signals (pins and peripherals)

There are **16 CLB input selection latches**. Each selects one signal — either an external pin (routed in through **CLBIN0PPS...CLBIN3PPS** / PPS) or an internal peripheral output — to bring into the fabric. These 16 signals are the connection to the rest of the chip.

2.2 Programmable edge detectors / input synchronizers

Each of the 16 selected inputs passes through a **programmable edge detector** before entering the fabric (the **CLB Input Synchronizer latches** configure it):

- **Positive edge** triggered (default),
- **Negative edge** triggered (set Input Synchronizer[0] = **1**),

- or **bypassed** entirely (signal fed in asynchronously, no synchronization).

If a BLE uses its output flop, take care that a *bypassed* (unsynchronized) input does not drive the flop into metastability.

See [§29.4.2 Programmable Edge Detectors](#).

2.3 Software inputs — CLBSWIN

User firmware can "bit-bang" values straight into the fabric through the 32-bit **CLBSWIN** register, formed from four 8-bit registers **CLBSWINU** : **CLBSWINH** : **CLBSWINM** : **CLBSWINL** (**CLBSWIN[31:0]**). Any of these 32 bits can be selected as a LUT input.

Write semantics (important):

- Write **CLBSWINU**, **CLBSWINH**, **CLBSWINM** **first**, then **CLBSWINL** **last**.
- Writing **CLBSWINL** latches all four bytes at once, sets the **BUSY** bit in **CLBCON**, and locks the registers for **one BLE_clk cycle** while they synchronize.

See [§29.4.1 CLB Software Input Register](#).

2.4 Internal feedback inputs

Two on-chip-generated signal groups are also available as inputs to the BLEs:

- the **3-bit counter** outputs (§3.2), and
- the **BLE outputs themselves**, fed back as possible inputs (so logic can be cascaded across BLEs).

3. Elements the CLB is built from

3.1 Basic Logic Elements (BLEs) — 32 of them

The **BLE is the primary building block**; the CLB module contains **32 BLEs**. Each BLE has two parts:

Element	Description
Look-Up Table (LUT)	A memory array with 4 input bits → 1 output bit , i.e. 16 storage elements (one per input combination). The 4 inputs act as the address selecting which stored bit is presented at the output. This implements any arbitrary 4-input combinational function.
Output flop	An optional flip-flop on the LUT output. The BLE Flop Select bit chooses whether the BLE output is the LUT result directly (combinational) or the registered/flopped version (sequential).

The LUT itself has **no reset**. The output flop is reset by: setting the **CLBMD** PMD bit, a device Reset, or clearing the **EN** bit. Unused LUTs should be configured to all-0 to minimize current.

Figure 29-1. Basic Logic Element (BLE) Simplified Block Diagram

See [§29.2 Basic Logic Element \(BLE\)](#).

3.2 Dedicated 3-bit hardware counter

For state-machine designs the CLB provides one **dedicated 3-bit counter**:

- **Clocked by** **BLE_clk** (the output of the CLB clock divider, §5).
- **Counter Stop** and **Counter Reset** can each be driven by **any of the 32 BLE outputs** (configurable).
- Each of the 3 count bits is available as an input to the BLEs, selected via the **Counter[n] Output Selection latches**.

 Figure 29-2. 3-Bit Counter Block Diagram

See [§29.3 Dedicated 3-Bit Counter](#).

3.3 Routing / Configuration latches

The interconnect is itself made of configuration latches, all written from the netlist by the NVM scanner:

- **CLB Input Selection** latches (which 16 signals enter, §2.1),
- **CLB Input Synchronizer** latches (edge-detector mode, §2.2),
- **BLE Input Selection** latches (which signals feed each BLE's 4 LUT inputs),
- **CLB Look-Up Table** latches (the 16 bits of each LUT),
- **BLE Flop Select** latches (combinational vs. registered output),
- **Counter Stop / Reset / Output Selection** latches (§3.2),
- **CLB Output Selection** latches (which BLE drives each PPS output, §4.1),
- **CLB Interrupt Selection** latches (§4.3),
- **CLB Clock Divider** latches (§5).

4. Outputs

4.1 Eight PPS outputs — **CLBPPSOUT[0..7]**

The CLB exposes **8 outputs**, **CLBPPSOUT[0]..CLBPPSOUT[7]**, that reach external pins through the **Peripheral Pin Select (PPS)** module. For each PPS output, the **CLB Output 'n' Selection** latches pick **one of four unique BLE outputs** to drive it.

 Figure 29-11. CLB Output 'n' Selections via the Configuration Interface

The **CLBPPSCONn** registers each hold two 4-bit **OESELn** fields, one per **CLBPPSOUT** signal. **OESELn** controls the pin's output-enable (TRIS): values **0000–0111** leave output-enable under **TRISx** control, while **1000–1111** tie the output-enable to a specific BLE output (**BLE[3]**, **BLE[7]**, ... **BLE[31]**) — this lets the CLB drive a **tri-state bus** (dynamically enable/disable the driver from logic).

OESELn	Output-enable source
1111	BLE[31]
1110	BLE[27]
1101	BLE[23]
1100	BLE[19]
1011	BLE[15]

OESSELn	Output-enable source
1010	BLE[11]
1001	BLE[7]
1000	BLE[3]
0111–0000	TRISx (normal pin control)

See §29.5 CLB Outputs and §29.5.1 CLB PPS Output Selections.

4.2 Internal peripheral connections

BLE outputs are also routed **internally** to other peripherals (no pin needed), and are fed back as possible BLE inputs. Per data-sheet Table 29-2:

Peripheral	Function driven by the CLB
ADC	ADC auto-conversion trigger source
CCP	CCP capture source
TMR0	TMR0 clock input
TMR1	TMR1 clock input / TMR1 gate source
TMR2	TMR2 clock input / TMR2 external reset source

4.3 Interrupts — four of them

The CLB provides **4 interrupts**. Each **CLB Interrupt Selection** latch picks one of the **32 BLE outputs** as the trigger; a **positive edge** on the selected output sets the corresponding **CLB1IFn** flag (in the PIR registers), and if **CLB1IEn** is set, an interrupt is generated.

Figure 29-12. CLB Interrupt Selections

See §29.7 CLB Interrupts.

5. Clocking

The CLB clock is selected with the **CLBCLK** register (**CLK[3:0]** field). The selected source passes through a **configurable clock divider** (set by the CLB Clock Divider latches in the netlist); the divider output is **BLE_clk**.

BLE_clk is the single synchronous clock of the whole fabric — it drives:

- all **32 BLE output flops**,
- the **dedicated 3-bit counter**, and
- the **CLBSWIN** register synchronization.

CLBCLK clock sources (Table 29-4)

CLK[3:0]	Clock source
0000	No clock selected
0001	CLBIN0PPS (external pin)
0010	CLBIN1PPS (external pin)
0011	CLBIN2PPS (external pin)
0100	CLBIN3PPS (external pin)
0101	FOSC (system clock)
0110	HFINTOSC
0111	LFINTOSC
1000	MFINTOSC (500 kHz)
1001	MFINTOSC (32 kHz)
1010	EXTOSC (external oscillator/crystal)
1011	ADCRC
1100	TMR0_overflow_OUT
1101	TMR1_overflow_OUT
1110	TMR2_postscaled_OUT
1111	Reserved

So the fabric can be clocked from the system clock, any internal oscillator, an external oscillator, a timer overflow/postscaler, the ADC RC clock, or one of four PPS input pins — then divided down to the desired `BLE_clk` rate.

Note: a fabric clocked from a **pin or peripheral** is asynchronous to `FOSC`. Using `FOSC/HFINTOSC` gives the fastest, CPU-synchronous fabric. For purely combinational designs (no flops, no counter) a clock is not strictly required.

See [§29.6 CLB Clock Selection](#) and [§29.9.6 CLBCLK](#).

6. Configuration & load sequence (summary)

From [§29.8 CLB Configuration](#):

1. In the Configuration Interface: select CLB inputs, edge-detector modes, BLE inputs, LUT contents, flop selects, interrupt sources, PPS output selections, counter stop/reset/output selections, and the clock divider.
2. The Configuration Interface writes the resulting **netlist into Program Memory**.
3. Select the clock source via `CLBCLK`; set the PPS output-enable via `CLBPPSCONn`.
4. Configure the **NVM Scanner** (with the CRC module) and run it to load the netlist into the CLB latches/LUTs.

5. Set the **EN** bit in **CLBCON** to enable the module.

Register	Role
CLBCON	EN (module enable), BUSY (CLBSWIN sync busy)
CLBCLK	CLK[3:0] clock source select
CLBSWINU/H/M/L	32-bit software input into the LUTs
CLBPPSCON1..4	OESELn output-enable select for CLBPPSOUT[0..7]

7. At a glance

Aspect	PIC16F13145 CLB
Logic elements	32 BLEs , each = 4-input LUT (16-bit) + optional output flop
Counter	1 dedicated 3-bit counter, clocked by BLE_clk
External/peripheral inputs	16 selectable, each with an edge detector / synchronizer
Software inputs	32-bit CLBWIN (bit-banged from firmware)
Feedback inputs	3 counter bits + BLE outputs
Outputs to pins	8 CLBPPSOUT[0..7] via PPS (each = 1 of 4 BLE outputs)
Internal outputs	ADC trigger, CCP capture, TMR0/1/2 clock/gate/reset
Interrupts	4 , each from any of the 32 BLE outputs (positive edge)
Clock	CLBCLK source → divider → BLE_clk (feeds all flops, counter, CLBSWIN)
Configuration	Netlist in PFM, loaded by the NVM scanner, then EN = 1